

F24-061-D-CrimeSight

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Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Multi Crime Classification and Detection: Client Desktop App . .	2
1.3.2	Crime Analytics: Client Desktop and Mobile App	3
1.3.3	Crime Reporting: Client Desktop and Mobile App	3
1.3.4	Suspicious Activity Tracking: Client Desktop App	3
1.3.5	Tracking Analytics: Client Desktop and Mobile App	4
1.3.6	Clip Sharing: Client Desktop and Mobile App	4
1.3.7	Remote Feed: Client Desktop and Mobile App	4
1.3.8	Emergency Contacts: Client Desktop and Mobile App	4
1.4	User Classes and Characteristics	4
2	Project Requirements	7
2.1	Use-case/Event Response Table/Storyboarding	7
2.2	Functional Requirements	15
2.2.1	Multi Crime Classification And Detection	15
2.2.2	Crime Analytics	15
2.2.3	Crime Reporting	16
2.2.4	Suspicious Activity	16
2.2.5	Tracking Analysis	16
2.2.6	Clip Sharing	17
2.2.7	Remote Feed	17
2.2.8	Emergency Contacts	17
2.3	Non-Functional Requirements	17
2.3.1	Reliability	17
2.3.2	Usability	18
2.3.3	Performance	18
2.3.4	Security	18
2.4	Domain Model	19
3	System Overview	21

3.1	Architectural Design	22
3.1.1	Initial Architectural Design	22
3.1.2	High Level Design Overview	22
3.1.2.1	Client Side	22
3.1.2.2	Server Side	22
3.1.2.3	Benefits of Client Server Architecture	22
3.1.2.4	Module Interactions	23
3.1.3	Detailed Architectural Design	24
3.2	Design Models	26
3.2.1	System level sequence diagrams:	26
3.2.2	Activity Diagrams:	30
3.2.3	Data Flow Diagram:	37
3.2.4	State Transition Diagram:	39
3.2.5	Data Representation Diagram:	41
3.2.5.1	Tables and Relationships:	42
3.2.5.2	Key Tables:	42
3.3	Data Design	42
3.3.1	CCTV Video Feed	42
3.3.2	Crime Model Output	43
3.3.3	User Personal Information	43
3.3.4	Data Storage Items	43
	References	45

List of Figures

2.1	Usecase Diagram	7
2.2	Domain Model	19
3.1	Initial Architecture Design	22
3.2	Data Flow Diagram	25
3.3	System Sequence Diagram for Viewing Camera Feed	26
3.4	System Sequence Diagram for Viewing Crime Analytics	26
3.5	System Sequence Diagram for Requesting Emergency Contacts	27
3.6	System Sequence Diagram for Requesting Person Tracking	27
3.7	System Sequence Diagram for Sharing Camera Feed	28
3.8	System Sequence Diagram for Managing Authorities	28
3.9	System Sequence Diagram for Receiving Alerts	29
3.10	Activity Diagram of View Camera Feed	30
3.11	Activity Diagram of View Crime Analytics	31
3.12	Activity Diagram of View Emergency Contacts	32
3.13	Activity Diagram of Person Tracking	33
3.14	Activity Diagram of Share Camera Feed	34
3.15	Activity Diagram of Manage Authorities	35
3.16	Activity Diagram of Recieve Alert	36
3.17	Data Flow Diagram	38
3.18	State Transition Diagram	40
3.19	Data representation diagram	41

List of Tables

2.1	Use Case: Receive Alerts	8
2.2	Use Case: Track Person	9
2.3	Use Case: View Camera Feed	10
2.4	Use Case: View Crime Analytics	11
2.5	Use Case: View Emergency Contacts	12
2.6	Use Case: Share Camera Feed	13
2.7	Use Case: Manage Authorities	14
2.8	Event Response Table	15

Chapter 1

Introduction

With Crime Sight, we want to revolutionize the way traditional security systems operate. Instead of reporting the crime manually after it has happened, we aim to detect and report the crime when it is happening live so that the relevant authorities can be alerted making it harder for the culprits to be able to escape. With the system in place, we aim to not only decrease the time to report a crime but also provide a platform where it is easier and convenient for users to share camera feeds, track people, and watch live feeds remotely from another device. [1].

1.1 Problem Statement

The current crime reporting system is reactive rather than proactive. Most of the crimes in today's society go unreported as they might be considered either too inconsequential or there is a lack of evidence for the law enforcing authorities to be able to take any tangible action just because they do not have access to cameras everywhere especially on private establishments. Apart from authorities, the work force cannot be present at all times to watch everything going on nor can anyone fight when security is compromised and their lives are in danger. As such, in all these cases, the culprits get away with their wrongdoings just because the current methods of reporting crime and gathering evidences require too many steps and are often too late. Often the crimes happen and are not reported on time which gives the offenders luxury of time to escape and timely action cannot be taken. These factors play part in rising crime rates.

1.2 Scope

The scope of our project includes detection and reporting of crimes within indoor environments. However, it can be extended to outdoor environments. In the case of extension, there is a possibility of reduction in accuracy due to external factors present in the outdoor environments that our product does not cater to. Our system will be engineered to detect a wide range of crimes including fight detection, weapon detection, detection of fire induced crimes and destruction of property. Our project includes a reporting system. It comprises of alerting the relevant authorities through Whats App and the user through the mobile application when a crime is detected. The alert for the user will be generated on the desktop as well as the mobile of the user. This makes sure user is not left in the dark regarding the crime taking place and is alerted at all costs. Our solution also provides a user friendly interface for easy viewing of the CCTV footage and performing relevant actions. Sharing of CCTV footage with other users of Crime Sight can easily be done through our product using our easy-to-navigate interface. Suspicious person tracking feature allows the user to track the culprit, aiding in capturing the responsible person by the relevant authorities. The desktop interface also provides a deep insight into the crimes that have taken place through an analytics portion. This section allows the user to have an overview of the previous crimes. It also allows the user to better understand on how to predict and prevent such crimes in the future.

1.3 Modules

1.3.1 Multi Crime Classification and Detection: Client Desktop App

In this module we are going to be working on building a model that is able to classify between 4 main crime categories: Fire Detection, Fight Detection, Weapon Detection, and Destruction of Property. [3][5]

1. Detect when a crime is happening or has happened.
2. Classify between 4 main crime types and normal scenarios.
3. Setup a Multi Camera Crime Handling System
4. The 4 main crime types include:
 - Fire Induced Crime
 - Fight Detection
 - Weapon Detection

- Destruction of Property

1.3.2 Crime Analytics: Client Desktop and Mobile App

Of all the crimes that happen within the user's setup, the Crime Analytics Module will provide a number of statistics on what happened on a day to day, month to month and year to year basis.

1. Types of Crimes being committed.
2. Timings on when crimes usually happens.
3. Number of crimes being committed.
4. Safety Advice based on the top crimes being committed.

1.3.3 Crime Reporting: Client Desktop and Mobile App

Automatic reporting of crimes to the relevant authorities and the user reporting space so that everyone is alerted in due time on what is going on alongside the video evidence of the crime.

1. Automatic Reporting to Relevant Authorities using Whats App.
2. Automatic Reporting to the Camera Setup owner using the Mobile Application.
3. Option to alert the organization security system through Whats App.
4. Options to choose alerts to authorities based on the type of crime.

1.3.4 Suspicious Activity Tracking: Client Desktop App

Be able to select an individual through the Desktop Application and trace back their steps compiling a feed of the actions and steps of the individual through the multiple cameras setup. [2]

1. Allow user to select an individual to track.
2. Allow user to set the time limit to track based on the past available camera feeds.
3. Provide user with the combined clips from multiple cameras of the steps taken by the individual.

1.3.5 Tracking Analytics: Client Desktop and Mobile App

Provide the user with relevant summary on what the tracked individual did.

1. Provide analytics relevant to the individual tracked on Desktop Application.
2. Provide analytics relevant to the individual tracked on Mobile Application.

1.3.6 Clip Sharing: Client Desktop and Mobile App

Allow users to be able to easily share camera feed clips with other users after they have made an account on Crime Sight.

1. Combine multiple clips from a single feed and share them with others.
2. Available users to specify the number of times the clip can be downloaded in order to protect the confidentiality of the clip.

1.3.7 Remote Feed: Client Desktop and Mobile App

Allow users to be able to monitor the camera feeds remotely on their mobile devices.

1. View Camera Feed on Mobile Application.
2. View Camera Feed on Desktop Application.

1.3.8 Emergency Contacts: Client Desktop and Mobile App

Provide users with relevant emergency contacts for relevant authorities,

1. Provide Emergency Contacts on Mobile Application.
2. Provide Emergency Contacts on Desktop Application.

1.4 User Classes and Characteristics

Following are the various user classes will use this system, and their pertinent characteristics.

User class		Description
Property Owner		A Property Owner can be defined as a person or establishment that occupies the property within which the Crime Sight system is set up. Currently, there are 1000 Property Owners and it is projected that 60 percent will make use of the systems on a daily basis. These users will mainly utilise the system for watching the activities within and around their properties and being alerted of any potential criminal events. Approximately 70 percent will use the system in desktop and mobile applications whereas 30 percent will use mobile appliances only. For new users of the system, only a short training period is necessary given the simplicity of the basic functions of the system.
Law Enforcement	En-	Law Enforcement Users are authorized users from different agencies who are alerted by the system and receive video evidence from the system. Law Enforcement users in various geographical regions will be integrated with the system. From the above system users survey, the system owners will mostly respond to unsolicited alarms and view videos to assist them in investigations. Little or no training is needed as their interaction is more focused by the use of complementary communication systems

User Classes and Characteristics

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

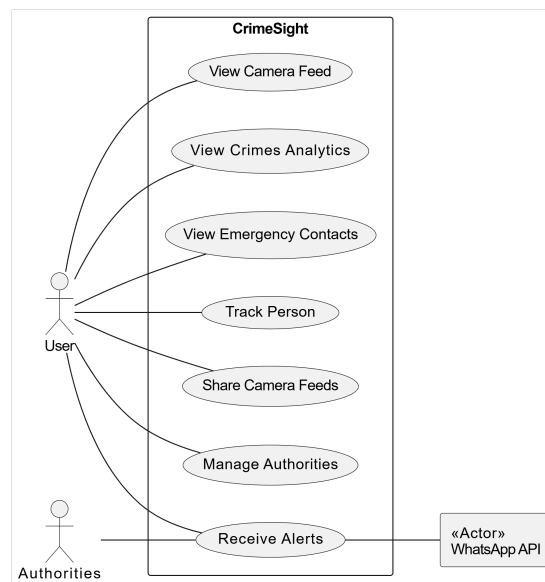


Figure 2.1: Usecase Diagram

Table 2.1: Use Case: Receive Alerts

Use Case ID	UC01
Use Case Name	Receive Alerts
Level	User Level Goal
Actors	User, Authorities
Type	Primary
Pre-conditions	Model detects crime.
Post-conditions	Crime Reported.
Frequency of Use	High
Description	System alerts the authorities and user. It logs the crime video.
Main Success Scenario	
Actor	System
	System detects a suspicious activity and automatically triggers an alert to authorities and user with the specific camera feed.
Authorities review the notification and decide on appropriate action.	
	Confirms that authorities have been notified and logs the status of the alert delivery. Logs the detected crime along with the video feed for later examination in the database.
User acknowledges the receipt of the alert.	
	Updates the crime status accordingly.

Table 2.2: Use Case: Track Person

Use Case ID	UC02
Use Case Name	Track Person
Level	User Level Goal
Actors	User
Type	Primary
Pre-conditions	A suspicious activity tracked.
Post-conditions	The offender has been tracked.
Frequency of Use	High
Description	Whenever a suspicious activity would be noticed, the system tracks the offender involved and their activity in all the previous footages of a certain time period.
Main Success Scenario	
Actor	System
	Upon detection, the system automatically begins tracking the offender, gathering data and monitoring their movements in real-time.
	It provides updates on the tracking progress, including key findings from the historical footage review, and alerts the user if any pertinent actions or decisions are required.
The user can make informed decisions regarding how to respond to the situation, such as alerting additional authorities, preparing to apprehend the suspect, or implementing safety protocols.	
	The system logs all relevant data and findings, ensuring a comprehensive record of the suspicious activity, the tracking process, and the conclusions drawn.

Table 2.3: Use Case: View Camera Feed

Use Case ID	UC03
Use Case Name	View Camera Feed
Level	User Level Goal
Actors	User
Type	Primary
Pre-conditions	-
Post-conditions	-
Frequency of Use	High
Description	User can view live camera feed from their device at all times.
Main Success Scenario	
Actor	System
Requests to view camera feed	
	Retrieves live feed from cameras
Selects certain camera feed	
	Retrieves feed from the selected camera.

Table 2.4: Use Case: View Crime Analytics

Use Case ID	UC04
Use Case Name	View Crime Analytics
Level	User Level Goal
Actors	User
Type	Secondary
Pre-conditions	-
Post-conditions	-
Frequency of Use	Low
Description	Crime Analytics Module will provide a number of statistics on what happened on a day to day, month to month and year to year basis.
Main Success Scenario	
Actor	System
Logs into the system and navigates to the Crime Analytics Module, accessing the relevant crime analytics page.	
	Retrieves and displays comprehensive crime data related to the user's setup. This includes a visual dashboard showcasing statistics on various crime metrics and highlights peak times when crimes usually occur, allowing the user to identify trends and patterns related to criminal activity and offers safety advice tailored to the current crime trends. This advice may include recommendations for preventative measures or community alerts.
reviews the displayed analytics and safety advice, gaining valuable insights into crime patterns that can inform their decision-making and strategy for improving safety.	
	The system updates the crime data regularly to ensure that the user always has access to the most current statistics and insights.

Table 2.5: Use Case: View Emergency Contacts

Use Case ID	UC05
Use Case Name	View Emergency Contacts
Level	User Level Goal
Actors	User
Type	Primary
Pre-conditions	User receives a crime alert.
Post-conditions	Crime Reported.
Frequency of Use	High
Description	Provide users with relevant emergency contacts for relevant authorities.
Main Success Scenario	
Actor	System
	Upon receiving the alert, the system automatically displays a list of pre-saved emergency contacts, which include relevant authorities such as local law enforcement, emergency services, and community safety contacts.
User selects the appropriate contact from the list (e.g., police, fire department, or emergency services) and initiates communication	
	Provides the user with the option to contact the selected emergency contact directly through the app, ensuring a seamless connection.

Table 2.6: Use Case: Share Camera Feed

Use Case ID	UC06
Use Case Name	Share Camera Feed
Level	User Level Goal
Actors	User
Type	Primary
Pre-conditions	-
Post-conditions	Feed shared with other users.
Frequency of Use	High
Description	Allow users to be able to easily share camera feed clips with other users after they have made an account on Crime Sight.
Main Success Scenario	
Actor	System
Navigates to the camera feed section.	
	Displays available clips recorded by their cameras.
Selects multiple clips from camera feed that they wish to share with other users.	
	Combines the selected clips into a single shareable video file, ensuring a seamless viewing experience for the recipients.
specifies the maximum number of times the combined clip can be downloaded by other users, setting restrictions to protect the clip's confidentiality then selects the recipients from their contact list or inputs email addresses to share the clip with.	
	Shares the clip and notifies the selected users that a clip has been shared with them, including the number of allowed downloads.

Table 2.7: Use Case: Manage Authorities

Use Case ID	UC07
Use Case Name	Manage Authorities
Level	User Level Goal
Actors	User, Authorities
Type	Primary
Pre-conditions	-
Post-conditions	Authorities are managed.
Frequency of Use	High
Description	System displays authorities to be selected in various domains. User adds or selects authority personnel.
Main Success Scenario	
Actor	System
Navigates to the section dedicated to managing authorities.	
	Displays a list of authorities categorized by various domains (e.g., police, fire department, emergency services) for the user to select from.
Selects the desired authority personnel from the list to manage and if necessary, the user can choose to add a new authority personnel by entering the required information (name, contact number, domain, etc.).	
	Validates the input information and saves the changes made by the user to the authority management database.

Table 2.8: Event Response Table

Event	System Response
Model detects crime	System alerts the authorities and user. It logs the crime video.
User views camera feed	System retrieves live feed from cameras.
User selects certain camera feed	System retrieves feed from the selected camera.
User views crime analytics	System displays relevant crime analytics page.
User views emergency contacts	System retrieves details and displays the emergency contacts list.
User tracks a person	System starts tracking and displays relevant details of the tracked person.
User shares camera feed	System displays camera feed logs to share. System displays list of users and sends the chosen video feed to the selected user.
User manages authorities	System displays authorities to be selected in various domains. User adds or selects authority personnel. System saves changes.

2.2 Functional Requirements

2.2.1 Multi Crime Classification And Detection

Following are the requirements for module 1:

1. FR1.1 The system shall detect and classify crimes into four categories: Fire Detection, Fight Detection, Weapon Detection and Destruction of Property.
2. FR1.2 The system shall distinguish between normal and crime scenarios.
3. FR1.3 The system shall process video feeds from multiple cameras simultaneously.

2.2.2 Crime Analytics

Following are the requirements for module 2:

1. FR2.1 The system shall display statistics of crime occurrences on day to day, month to month and year to year basis.
2. FR2.2 The system shall generate patterns in crime timings, types and frequency .

3. FR2.3 The system shall provide safety advice based on the top crimes being committed.

2.2.3 Crime Reporting

Following are the requirements for module 3:

1. FR3.1 The system shall automatically report relevant authorities using Whats App.
2. FR3.2 The system shall automatically report Camera Setup owner using Mobile Application.
3. FR3.3 The system shall have options to choose alerts to authorities based on the type of crime.
4. FR3.4 All crime reports shall include video evidence.

2.2.4 Suspicious Activity

Following are the requirements for module 4:

1. FR4.1 The system shall allow user to select an individual to through the Desktop Application and track them through the multiple cameras setup.
2. FR4.2 The system shall allow user to set the time limit to track based on the past available camera feeds.
3. FR4.3 The system shall provide user with the combined clips from multiple cameras of the steps taken by the individual.

2.2.5 Tracking Analysis

Following are the requirements for module 5:

1. FR5.1 The system shall provide the user with relevant summary on what the tracked individual did.
2. FR5.2 The system shall provide analytics relevant to the individual tracked on both Desktop and Mobile Application.

2.2.6 Clip Sharing

Following are the requirements for module 6:

1. FR6.1 The system shall allow user to combine multiple clips from a single feed..
2. FR6.2 The system shall allow users to share camera feeds with other users.
3. FR6.3 The system allow user to specify the number of times the shared clips can be downloaded by the other users.

2.2.7 Remote Feed

Following are the requirements for module 7:

1. FR7.1 The system shall allow the users real-time access to camera feeds remotely on both Mobile and Desktop Applications.
2. FR7.2 The system shall allow user to simultaneously view feeds from multiple cameras.

2.2.8 Emergency Contacts

Following are the requirements for module 8:

1. FR8.1 The shall provide user with relevant emergency contacts for relevant authorities.
2. FR8.2 The system shall allow the user to customize emergency contact lists.
3. FR8.3 The system shall provide user with immediate and easy access to contact emergency services from both Desktop and Mobile Applications.

2.3 Non-Functional Requirements

2.3.1 Reliability

REL-1: The system would be highly reliable, making sure any discrepancies in the model are covered up by the system's post processing.

REL-2: The system would achieve an average MTBF of 600 hours.

REL-3: The system would manage any crashes, or failures by sending proper crash logs, error messages, and try it's best to get back up within 5 minutes.

2.3.2 Usability

USE-1: The camera feeds will be available to user on the home page requiring no extra button clicks.

USE-2: Pressing on any Emergency Contact will result in a call being made directly without any extra clicks.

USE-3: The system will be user friendly making sure new users find it easier to navigate.

2.3.3 Performance

PER-1: The system shall classify activity type for 2 cameras simultaneously within a second.

PER-2: Alerts will be notified to relevant authorities within 5 seconds of finalizing crime detection and evidence generation.

2.3.4 Security

SEC-1: The application will use industry standard encryption algorithms for all personal user data and security feeds transmission.

SEC-2: The system will secure the users using two factor authentication.

SEC-3: The system will make sure the shared clips are only available for a specific period of time.

SEC-4: The system will manage a log of all audits.

2.4 Domain Model

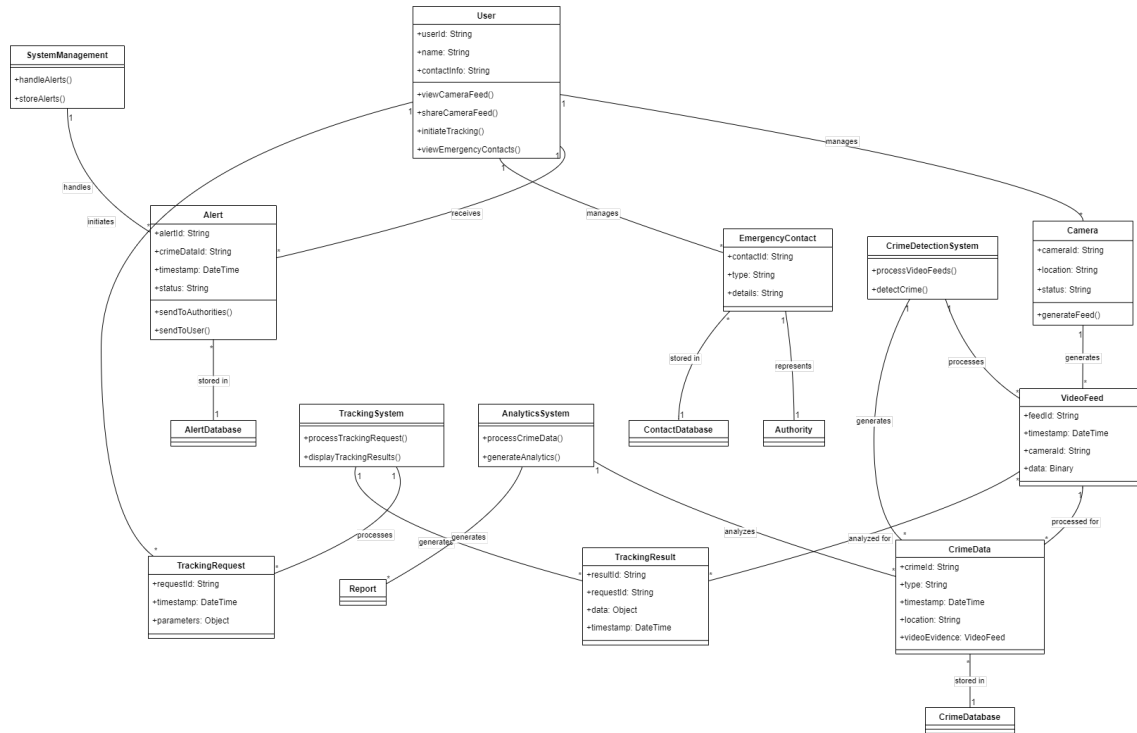


Figure 2.2: Domain Model

Chapter 3

System Overview

Crime Sight is an application that provides user with the ease of mind that their belongings are always safe, under watch and if anything was to happen to them, it would be instantly reported. Crime Sight continuously scans the scene for any abnormal crime activity that may lead to harm or damage to goods.

Crime Sight employs machine learning models to scan your surroundings that are being monitored by CCTV Cameras and make sure that everything is in order. This is made possible by predicting on the frames in the video by extracting useful features from them and processing them alongside other frames to be able to come at a conclusion on what is happening in real time. [3] [6] If any abnormal crimes are detected, Crime Sight will automatically notify all relevant users and authorities to make sure it does not take time for news to spread and for necessary action to be taken.

Crime Sight also provides the user with the option to be able to track individuals in past stored video feeds of the CCTV Cameras. It does this job by tracking people through their faces and assigning unique IDs to them so that they can be tracked across multiple cameras. [4] [2] In the end, the tracking module provides you with the combined clips of the tracked person, providing a complete and comprehensive solution in case the activity of a person is to be studied.

Moreover, Crime Sight also provides functionalities such as remote feed viewing from another device, clip editing and sharing with others, one tap emergency contacts and detailed analytics of both crimes and tracking. It gives the user more control over their CCTV devices while also providing a very easy to use user interface allowing the users to fully enjoy the services on their devices with ease.

3.1 Architectural Design

3.1.1 Initial Architectural Design

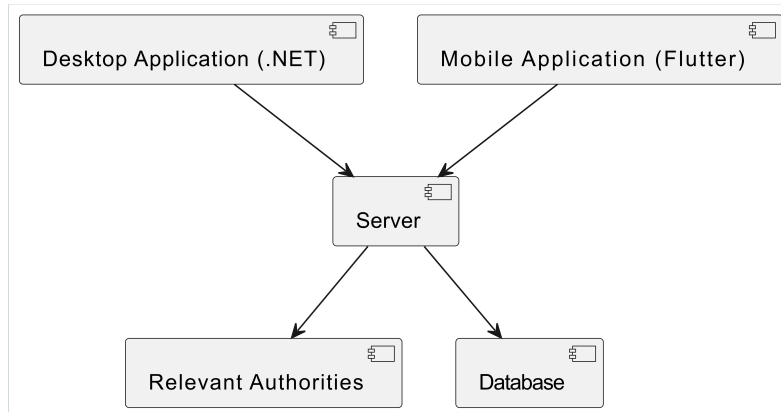


Figure 3.1: Initial Architecture Design

3.1.2 High Level Design Overview

The Architecture of our Application is Client Server where the User (Client) interacts using the Desktop and Mobile Application whereas the Server manages all the Databases, Tracking Models, Central Video Storage and Reporting to the relevant authorities.

3.1.2.1 Client Side

Manages all operations that can be easily ran on the Client's System. It consists of both the UI for Desktop and Mobile Application alongside the implementation logic of some important local features such as Crime Model, Clip Sharing, Local CCTV Footage storing etc.

3.1.2.2 Server Side

Manages all operations that require careful maintenance and high level security such as personnel data of users, high compute tasks such as the tracking module or effectively handling reporting to the Police Authorities to make sure they do not get spammed.

3.1.2.3 Benefits of Client Server Architecture

1. Scalability

2. Maintainability
3. Security
4. Resource Management

3.1.2.4 Module Interactions

1. **Crime Classification Module:** Uses frames from the CCTV Cameras to process the video and perform inference on the type of actions being committed.
2. **Crime Analytics Module:** Provides useful statistics and results based on the data generated by the Crime Classification Module
3. **Crime Reporting Module:** Reports to the user/owner and relevant authorities to make an early report of the crime being committed when the crime has been detected and approved by the Crime Classification Module.
4. **Suspicious Activity Tracking Module:** Allows users to be able to track selected individual through past footage.
5. **Tracking Analytics Module:** Uses the results generated by the Suspicious Activity Tracking Module and show them to the user.
6. **Remote Feed Module:** Allows user to view their camera feeds from other devices.
7. **Clip Sharing Module:** Allows user to share clips from the CCTV Camera Feeds with other users.
8. **Emergency Contact Module:** Allows user to be able to contact emergency services directly easily and faster.

3.1.3 Detailed Architectural Design

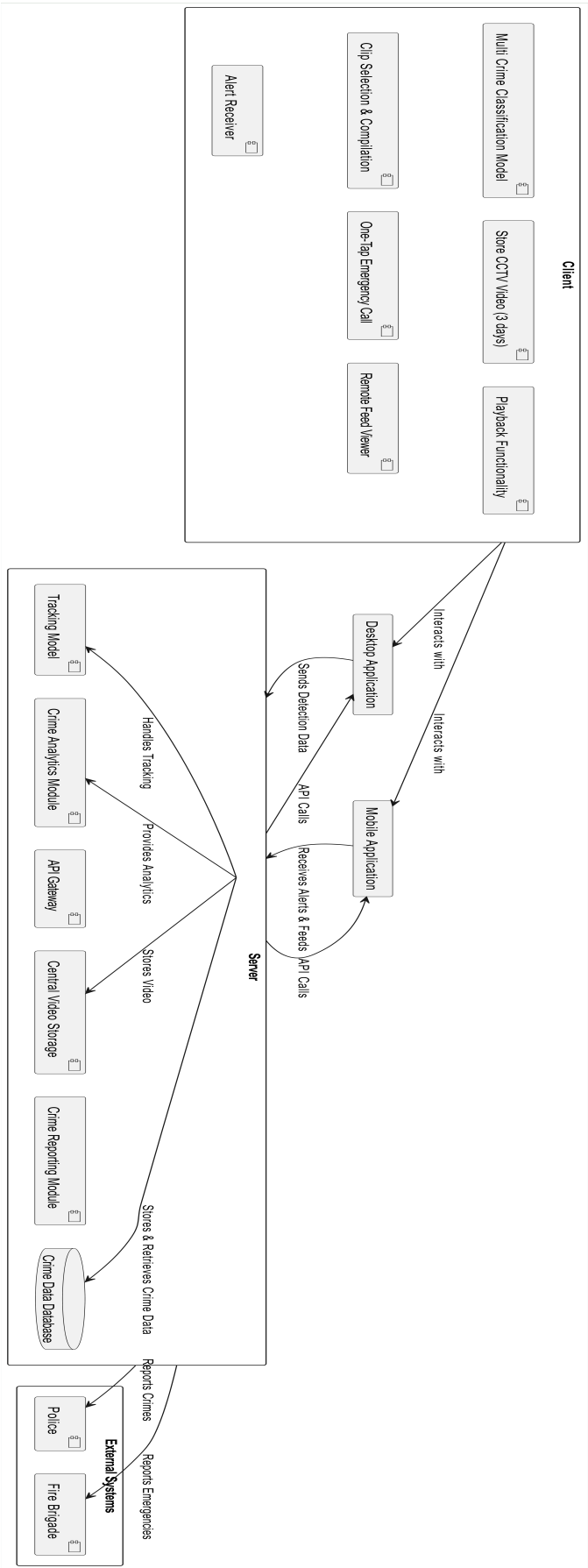


Figure 3.2: Data Flow Diagram

3.2 Design Models

Design Models for Procedural Approach

3.2.1 System level sequence diagrams:

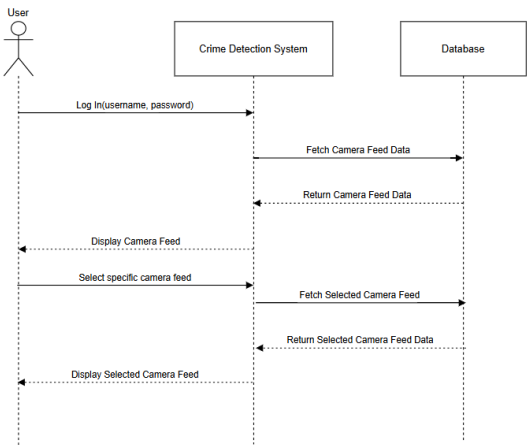


Figure 3.3: System Sequence Diagram for Viewing Camera Feed

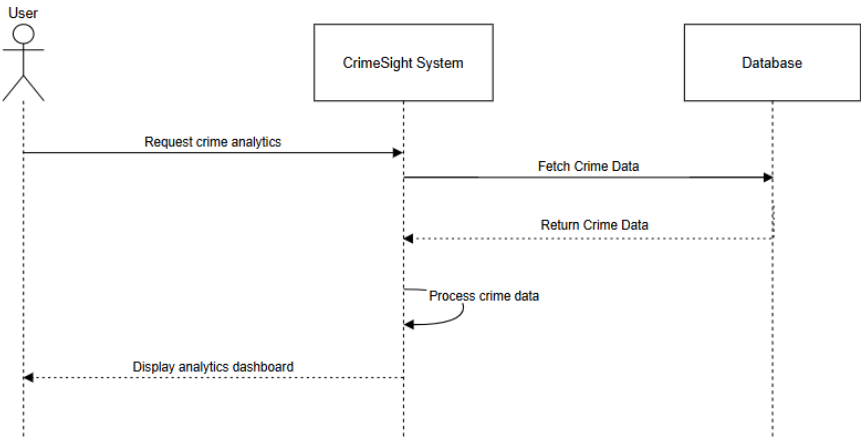


Figure 3.4: System Sequence Diagram for Viewing Crime Analytics

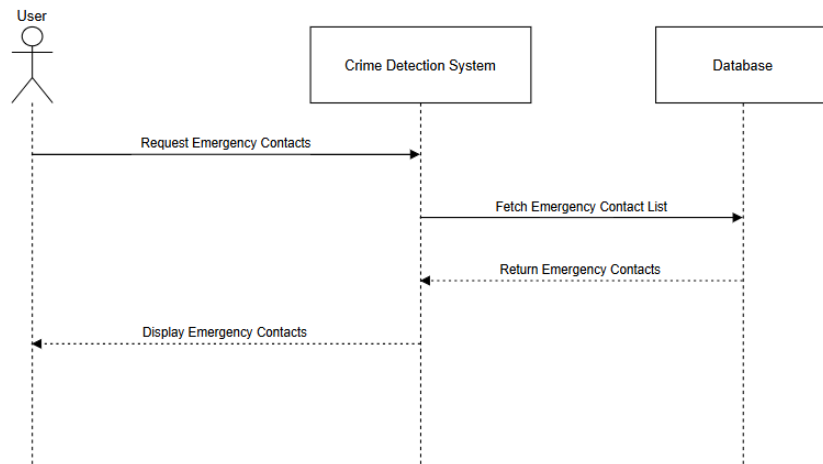


Figure 3.5: System Sequence Diagram for Requesting Emergency Contacts

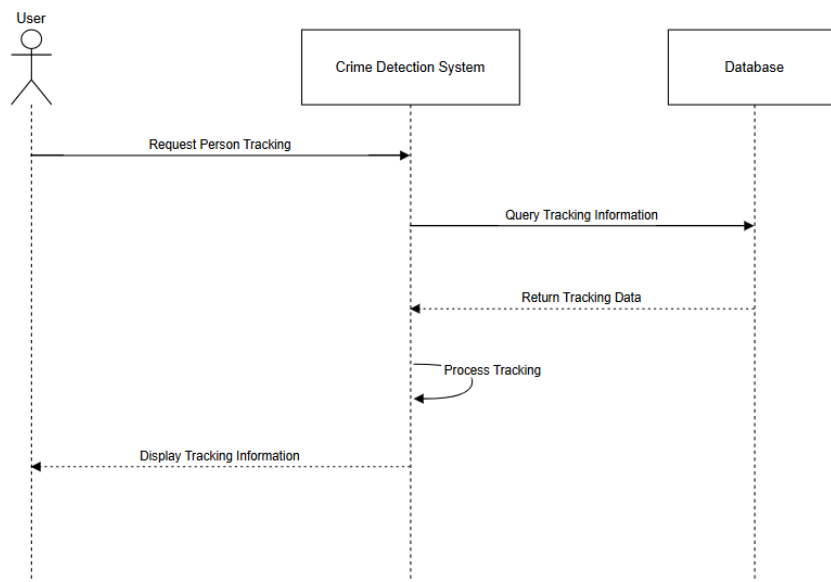


Figure 3.6: System Sequence Diagram for Requesting Person Tracking

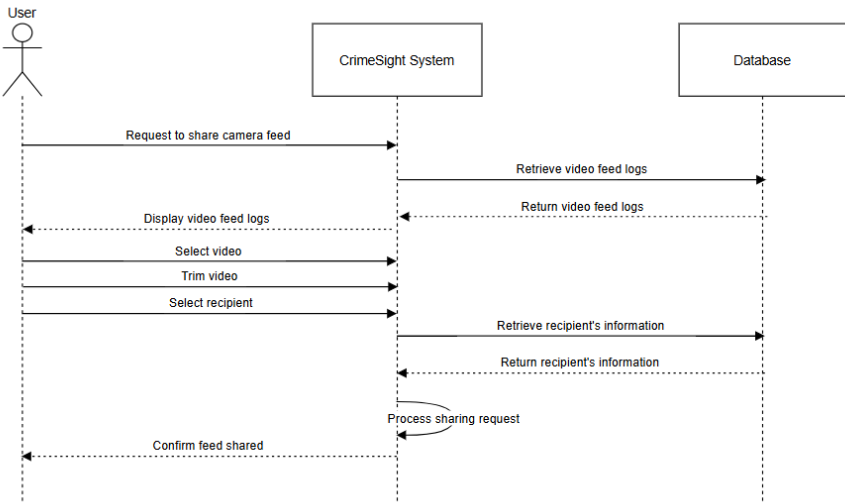


Figure 3.7: System Sequence Diagram for Sharing Camera Feed

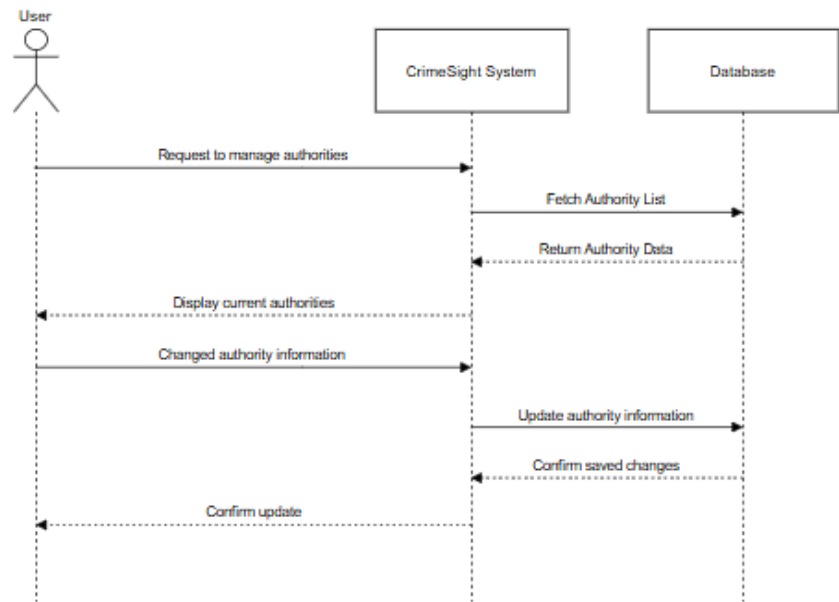


Figure 3.8: System Sequence Diagram for Managing Authorities

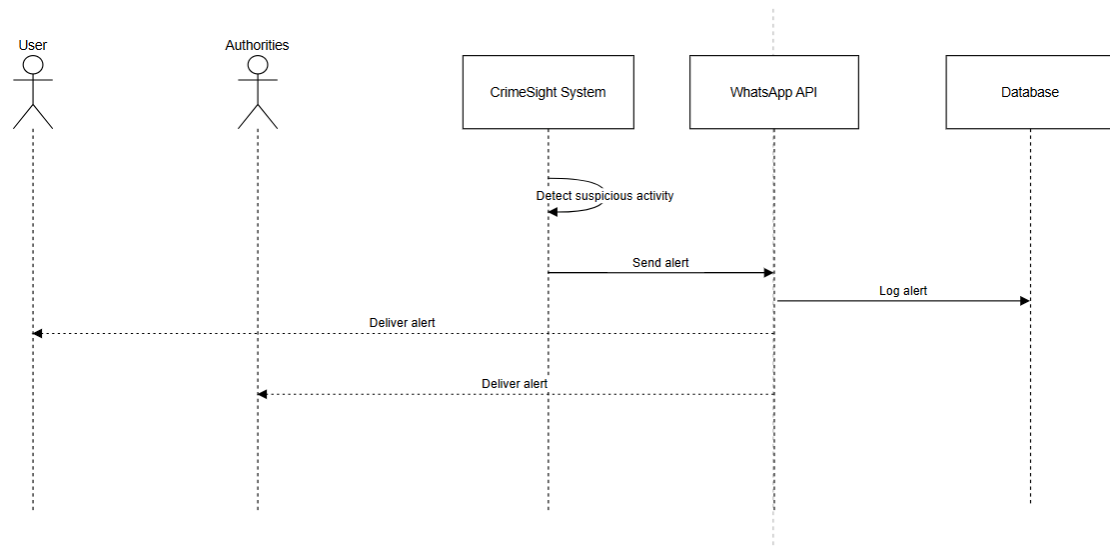


Figure 3.9: System Sequence Diagram for Receiving Alerts

3.2.2 Activity Diagrams:

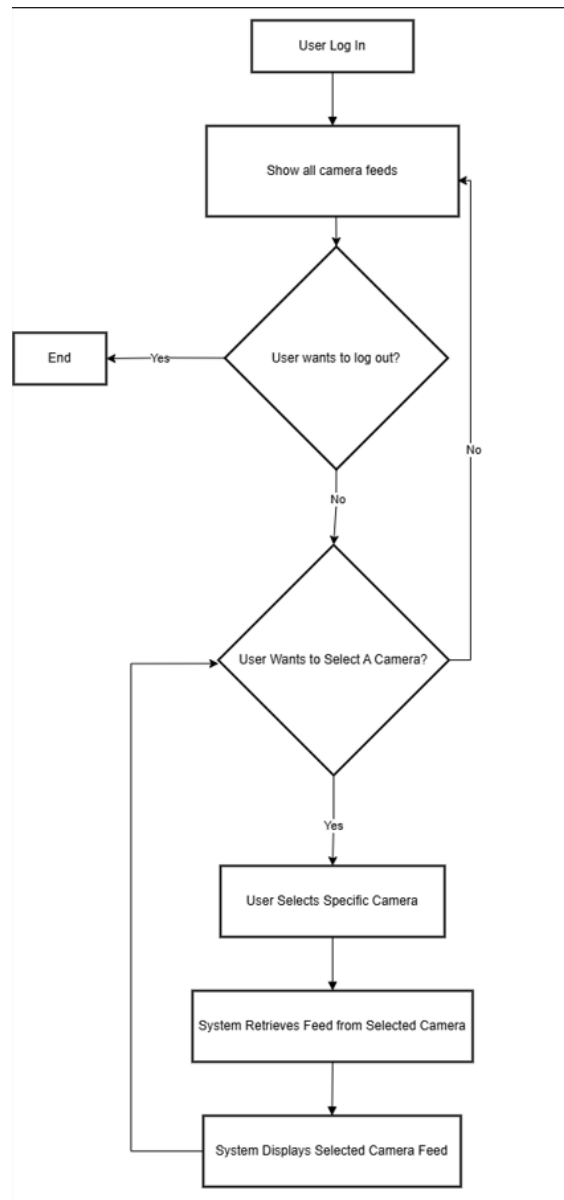


Figure 3.10: Activity Diagram of View Camera Feed

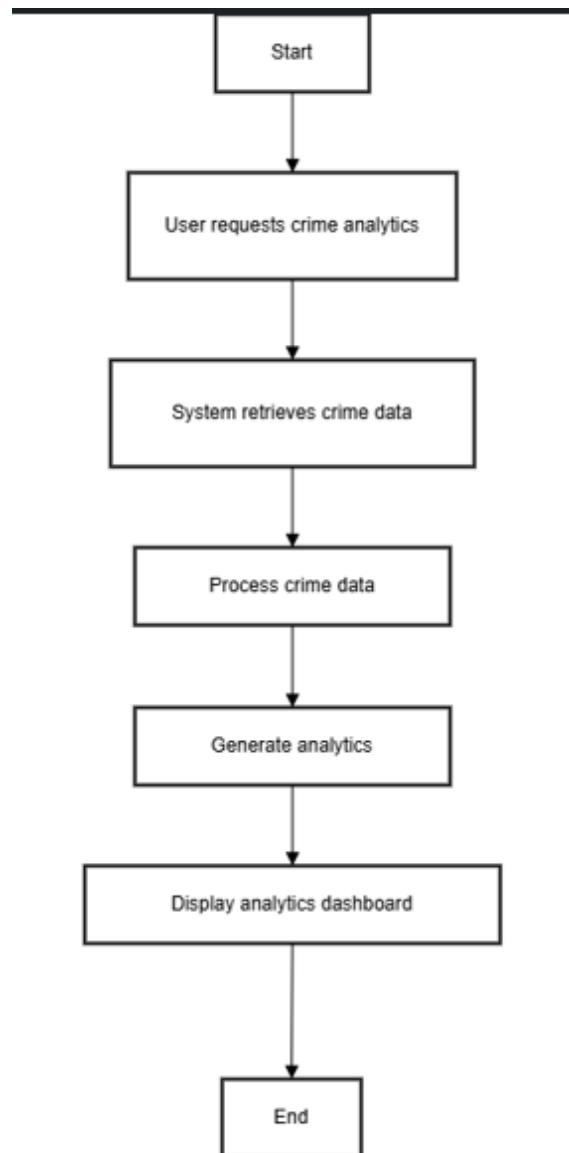


Figure 3.11: Activity Diagram of View Crime Analytics

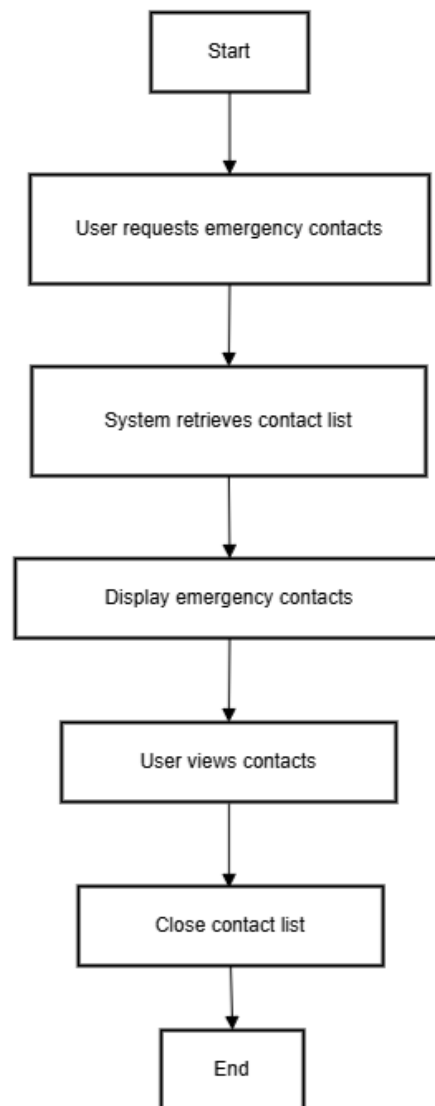
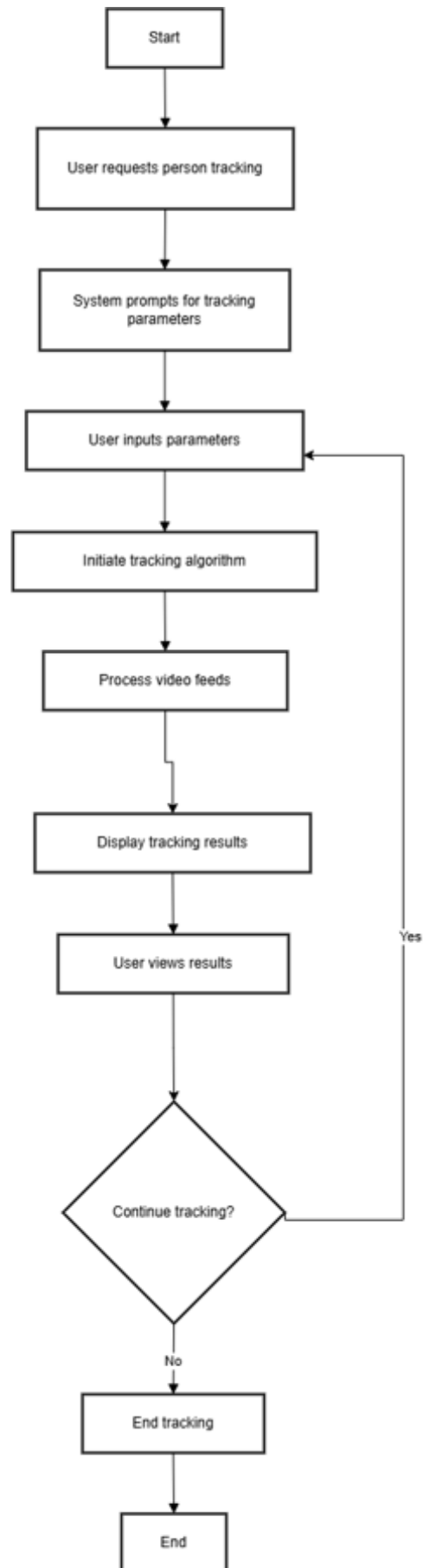


Figure 3.12: Activity Diagram of View Emergency Contacts



33
Figure 3.13: Activity Diagram of Person Tracking

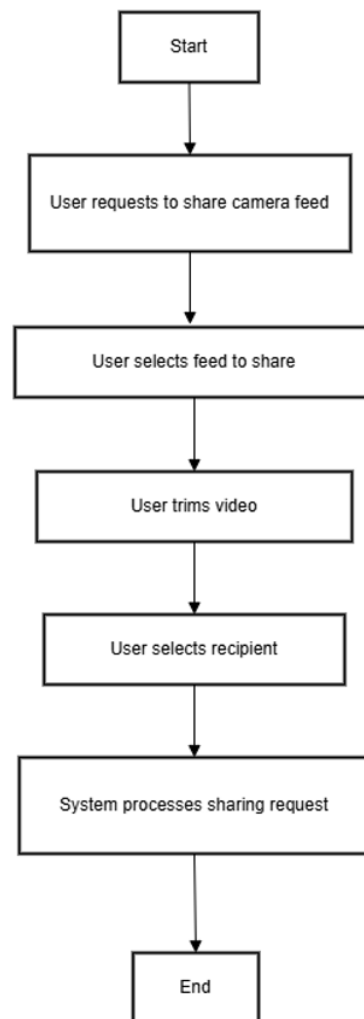


Figure 3.14: Activity Diagram of Share Camera Feed

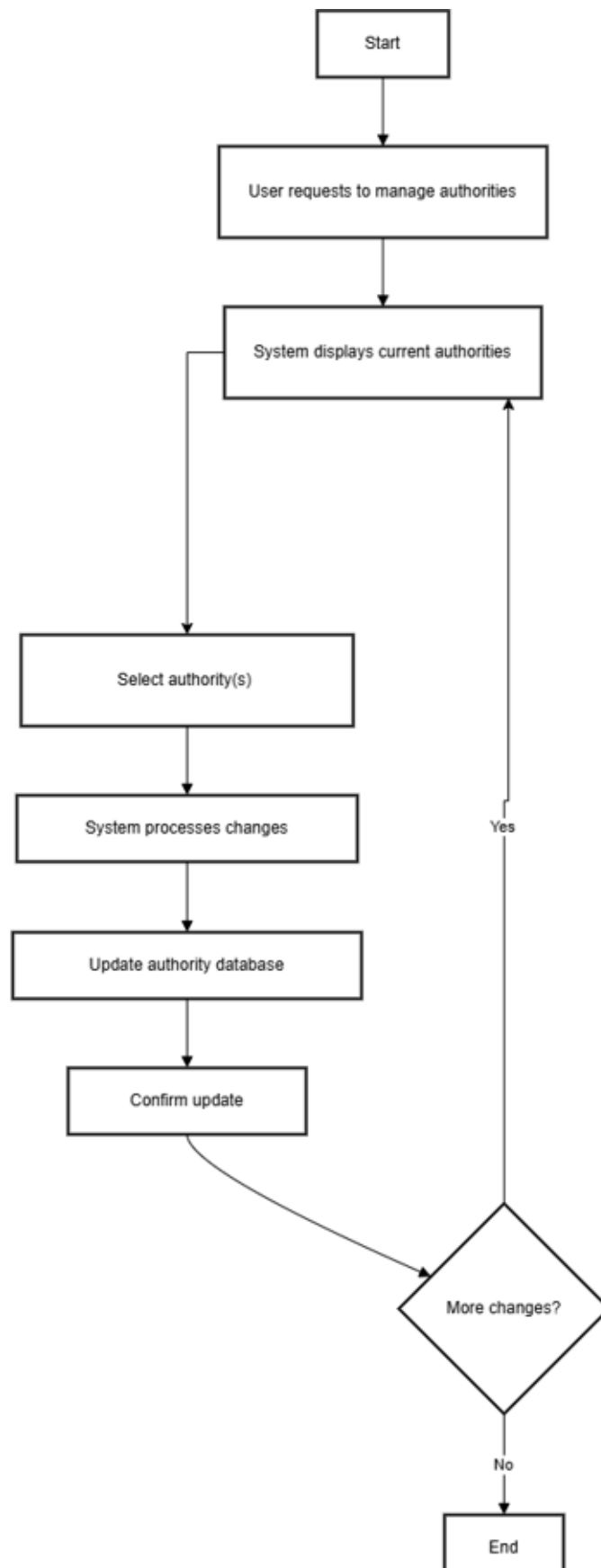


Figure 3.15: Activity Diagram of Manage Authorities

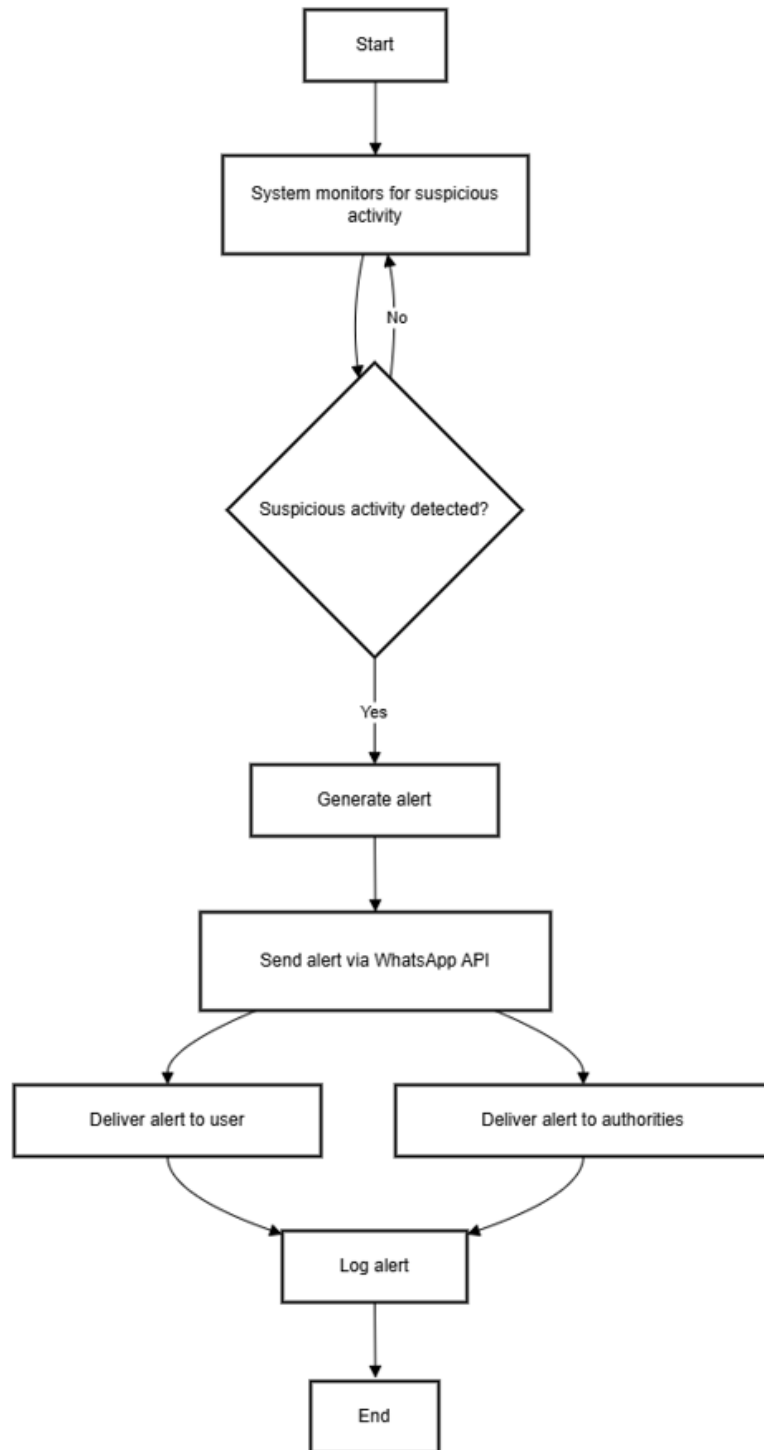


Figure 3.16: Activity Diagram of Recieve Alert

3.2.3 Data Flow Diagram:

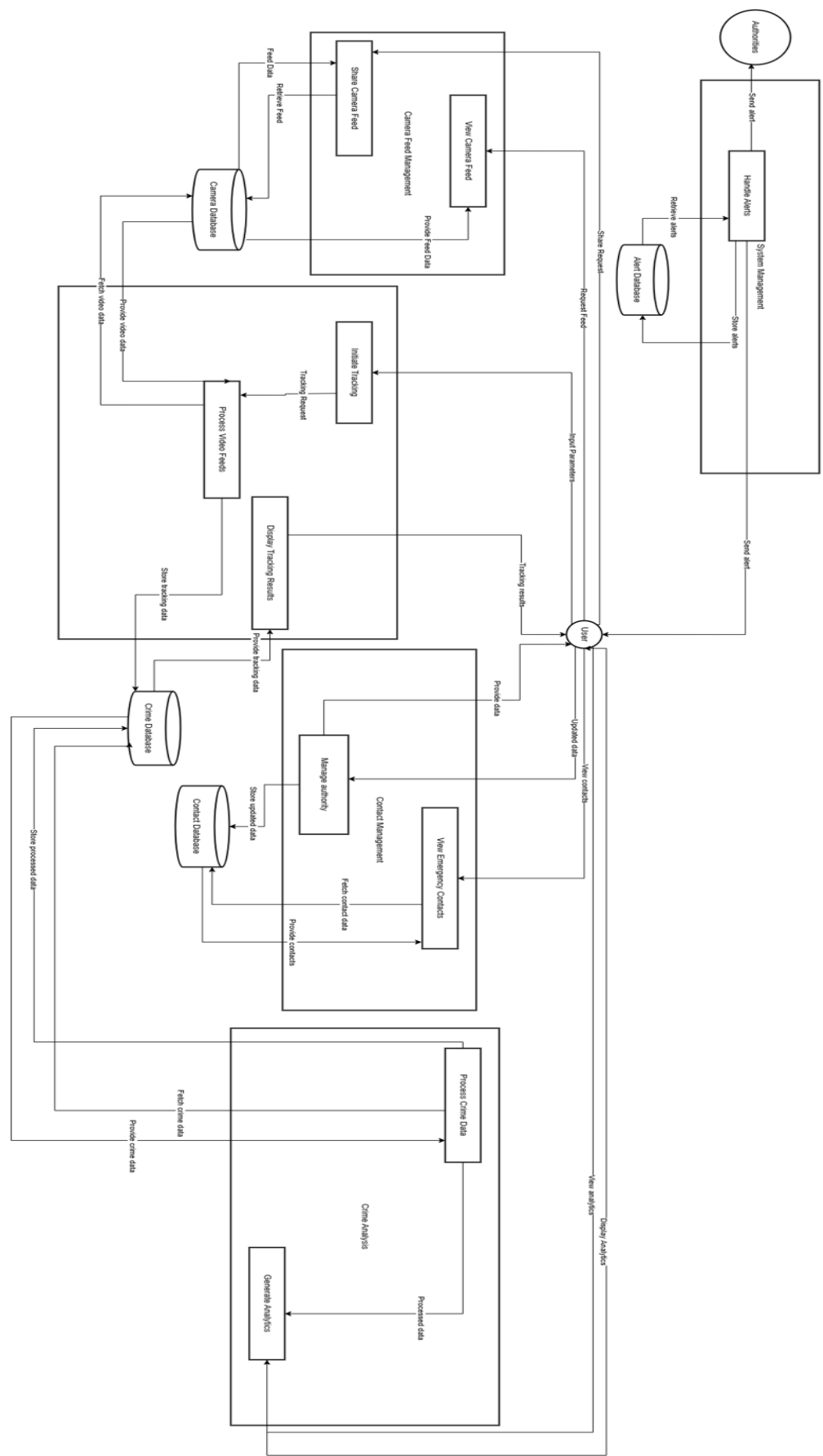


Figure 3.17: Data Flow Diagram

3.2.4 State Transition Diagram:

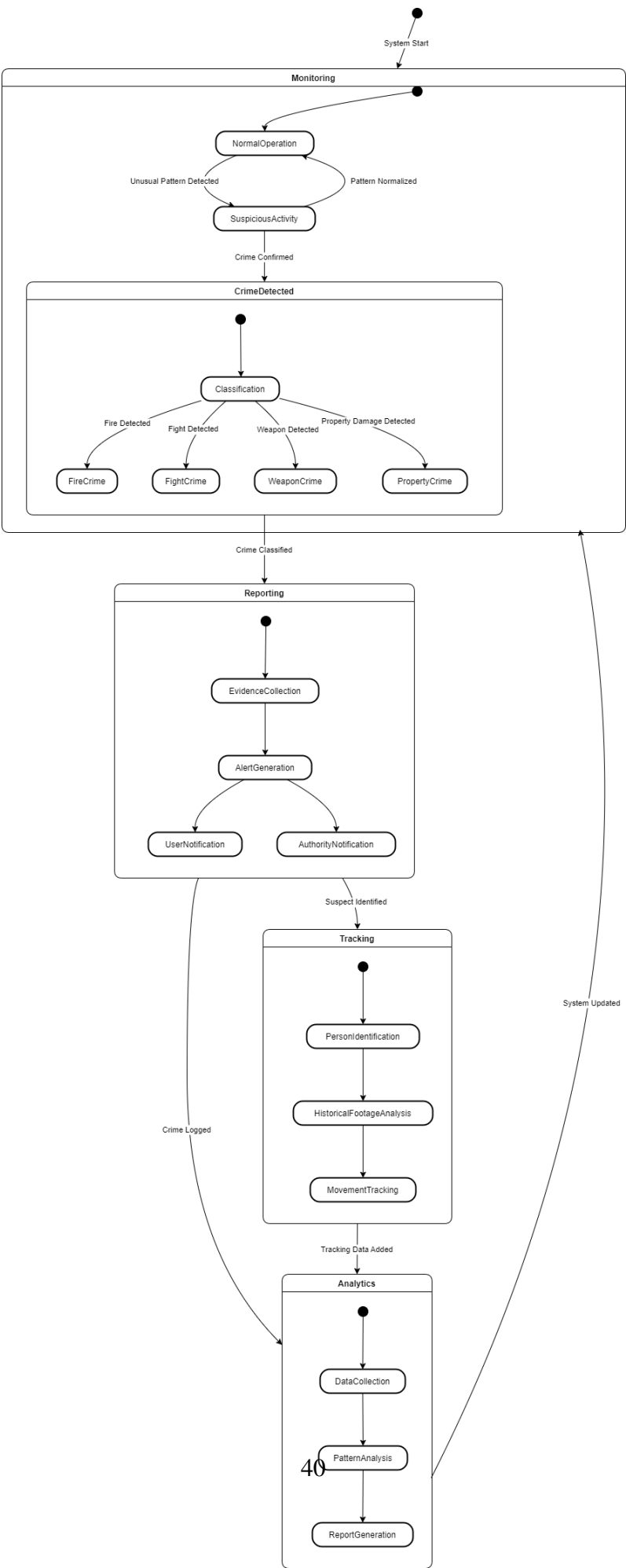


Figure 3.18: State Transition Diagram

3.2.5 Data Representation Diagram:

The following database schema represents our crime detection and monitoring system. It is composed of several interconnected tables.

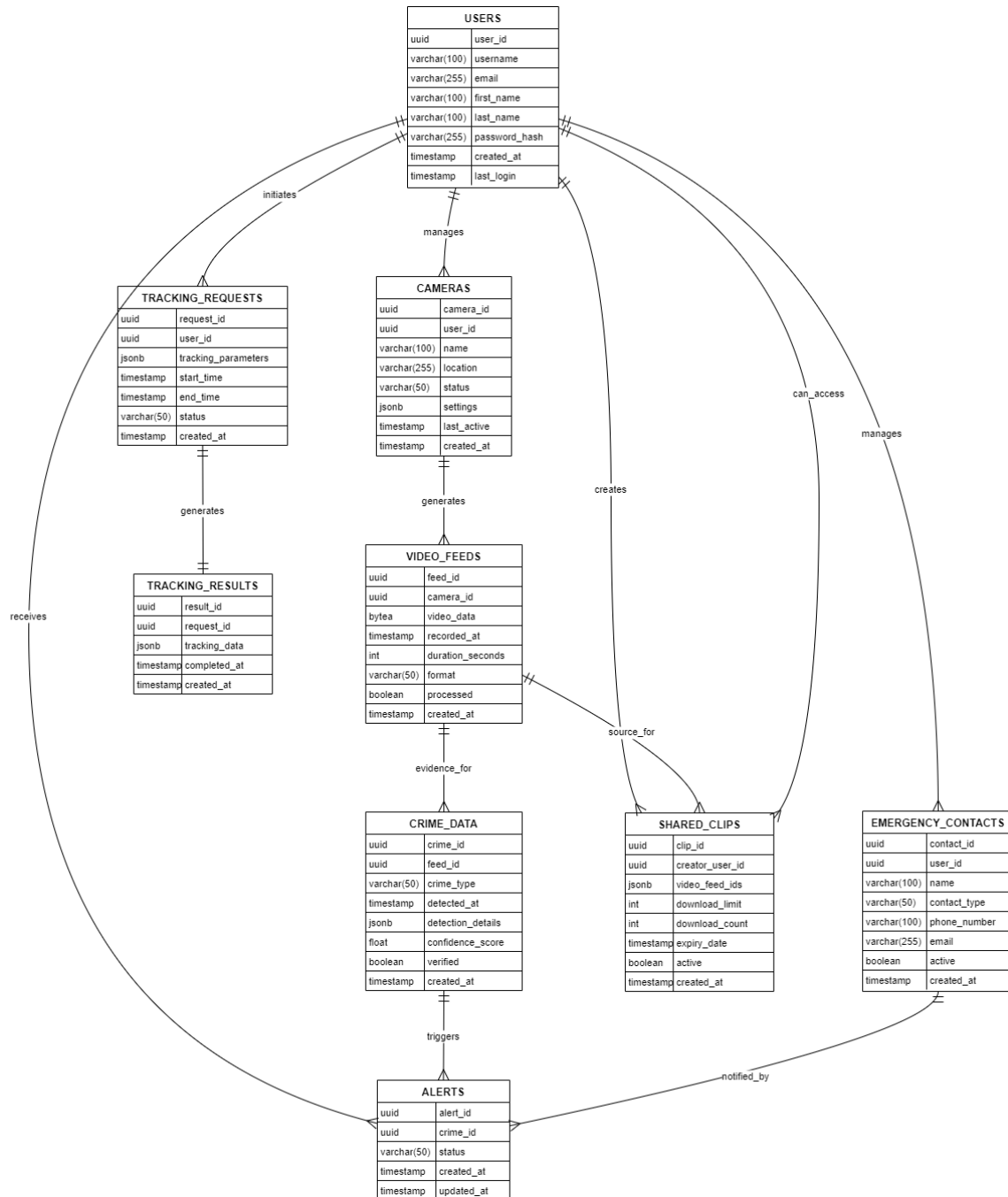


Figure 3.19: Data representation diagram

3.2.5.1 Tables and Relationships:

- **Primary Keys (PK):** Each table has a UUID as its primary key (K1).
- **Foreign Keys (FK):** The tables are connected via foreign keys using certain parameters.
- **Relationships:** These are represented with the help of crow's foot association.

3.2.5.2 Key Tables:

- **USERS:** Main table which contains information about the users. The username and the email have unique constraints.
- **CAMERAS:** Associated with the users, holds information about the cameras.
- **VIDEO_FEEDS:** Handles data related to CCTV video feeds.
- **TRACKING_REQUESTS:** Logs requests initiated by users to track specific activities, storing tracking parameters, start and end times, and created timestamps.
- **TRACKING_RESULTS:** Stores the results of tracking requests, including tracking data, timestamps, and completed times.
- **CRIME_DATA:** Captures detected crimes from video feeds, storing crime types, detection details, confidence scores, and verification statuses.
- **ALERTS:** Stores alerts triggered by detected crimes, tracking alert statuses, crime IDs, and alert creation and update times.
- **SHARED_CLIPS:** Manages video clips shared by users, including video feed IDs, download counts, expiration details, and active statuses.
- **EMERGENCY_CONTACTS:** Contains emergency contact details for users, including phone numbers and contact statuses.

3.3 Data Design

3.3.1 CCTV Video Feed

- The video stream is loaded in, and individual frames are extracted from the video.
- Frames are extracted at a fixed interval of every 10th successive frame.

- Once the frames are extracted, they are processed and sent to the crime classifier model for inference.
- The frames are discarded after the model is done with inference, depending upon the output.

3.3.2 Crime Model Output

- The probabilities generated by the model are processed to determine the type of action detected.
- The probabilities are then used to determine whether to alert the relevant authorities.
- The frames are converted back into video format to be used as evidence and sent to the owner and relevant authorities.
- The output at the end is discarded.

3.3.3 User Personal Information

- User data is stored securely in a database and can only be accessed by the Database Administrator or the user.
- Data is used by the application to perform actions such as logging in, sharing videos, and reporting to relevant authorities.

3.3.4 Data Storage Items

- **PostgreSQL** will be used to store the private information of the users.
- A **Video Sequence Generator** is used during the model training and testing process. It is responsible for handling all data related to model training.

Bibliography

- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.
- [2] Roberto Marroquin, Julien Dubois, and Christophe Nicolle. Wisenet: An indoor multi-camera multi-space dataset with contextual information and annotations for people detection and tracking. *Data in Brief*, 2019.
- [3] Waqas Sultani, Chen Chen, and Mubarak Shah. Real-world anomaly detection in surveillance videos. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018.
- [4] Nicolai Wojke, Alex Bewley, and Dietrich Paulus. Simple online and realtime tracking with a deep association metric, 2017.
- [5] Peng Wu, jing Liu, Yujia Shi, Yujia Sun, Fangtao Shao, Zhaoyang Wu, and Zhiwei Yang. Not only look, but also listen: Learning multimodal violence detection under weak supervision. In *European Conference on Computer Vision (ECCV)*, 2020.
- [6] Yixuan Zhou, Yi Qu, Xing Xu, Fumin Shen, Jingkuan Song, and Hengtao Shen. Batchnorm-based weakly supervised video anomaly detection, 2023.